SOLARRECOVER AI

Vendor-Neutral Solar Asset Intelligence & Digital Twin Platform

PRODUCT BRIEF

Turning fragmented PV operational data into explainable energy-loss diagnosis, financial impact analysis, recovery simulation, and actionable maintenance decisions.

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This brief defines the product vision, problem, users, differentiated value proposition, core capabilities, architecture, MVP scope, AI strategy, security principles, success metrics, and product roadmap for SolarRecover AI.

1. Executive Summary

SolarRecover AI is a software platform designed to create a continuously updated digital representation of photovoltaic assets and use physics-informed analytics, machine learning, and AI to identify abnormal performance, investigate probable causes, quantify energy and financial losses, simulate recovery scenarios, and guide maintenance teams toward evidence-based actions.

The product is deliberately positioned between raw monitoring and operational decision-making. It is not intended to replace established PV design/simulation tools, SCADA systems, or OEM monitoring platforms. Its central business objective is to answer a more operational question: where is energy being lost, why is it being lost, what is the impact, and what action could recover it?

The product thesis is supported by the supplied 2026 systematic review of photovoltaic Digital Twins. That review evaluated 61 peer-reviewed PVDT studies and found that most reported implementations lacked important DT characteristics, particularly bidirectional data flow and self-adaptability; only 3.3% met all the proposed eligibility criteria. filecite turn5file2 L174-L188

The same review recommends treating the Digital Twin as the continuously accurate digital counterpart of the real PV system, with prediction and diagnostic algorithms operating as additional layers. It also identifies multi-source intelligent analytics, actionable guidance, scalability, transferability, synchronization, latency, and cybersecurity as important future directions.
filecite turn5file3 L240-L284

Product thesis

SolarRecover AI proposes a differentiated combination of vendor-neutral PV Digital Twin + performance modelling + AI diagnosis + energy-loss attribution + financial impact + recovery simulation + maintenance feedback. This is a product/research differentiation hypothesis, not a claim that no existing system contains any of these individual capabilities.

One-line product definition

“SolarRecover AI is a vendor-neutral AI Digital Twin platform that transforms PV operational data into explainable energy-loss diagnosis, financial impact analysis, recovery simulation, and actionable maintenance decisions.”

2. Product Vision & Mission

Vision

Create an intelligent digital counterpart for solar assets that continuously understands their operational condition, explains energy losses, predicts future problems, and helps teams recover lost generation.

Mission

- Continuously model the operational state of PV assets.
- Establish expected performance under current environmental and operating conditions.
- Detect meaningful deviations from expected behaviour.
- Identify and explain probable causes using multiple evidence sources.
- Quantify estimated energy loss and its financial impact.
- Evaluate corrective-action scenarios before resources are committed.
- Prioritize maintenance using transparent, business-relevant factors.

• Capture repair outcomes so the system can improve over time.

3. Problem Definition

Industry problem

PV systems generate large quantities of operational and environmental data, but the information needed to make maintenance decisions can remain fragmented across inverter systems, SCADA, weather sources, inspection records, spreadsheets, and field-service workflows. The supplied review notes that PV monitoring infrastructure can be limited, particularly in distributed PV, creating both technical and economic challenges for robust DT solutions. It also identifies data integration and interoperability as areas requiring further development. □filecite□turn5file5□L370-L387□

Decision gap

A conventional alert answers “something is abnormal.” A high-value operational system should additionally support “why, how much, what happens if we act, and what should the team investigate?”

Current workflow	Decision gap	SolarRecover response
Monitoring	Abnormality without context	Contextual anomaly analysis
Engineering analysis	Manual correlation of multiple signals	Multi-source evidence synthesis
Maintenance planning	Many issues competing for attention	Impact-aware prioritization
Repair	Action separated from analytics	Technician guidance and work records
Post-repair	Outcome often not fed into analytics	Feedback loop into asset history and models

4. Target Users & Jobs-to-be-Done

User	Primary job	Key question
Plant Owner	Protect generation and revenue	How much energy/revenue is at risk?
Asset Manager	Manage portfolio health	Which assets need attention?
O&M; Manager	Allocate maintenance resources	What should the team work on first?
Solar Engineer	Investigate technical anomalies	What is the most likely cause and evidence?
Field Technician	Inspect and resolve issues	What should I check on site?
EPC / Service Provider	Monitor delivered assets	Where is performance diverging from expectations?
Analyst / Researcher	Study performance and models	What patterns explain the deviation?

Primary personas

1. O&M; Manager: needs a portfolio-level view that converts technical anomalies into prioritized work.

2. Solar Engineer: needs evidence, time-series context, asset relationships, and model outputs for diagnosis.

3. Field Technician: needs concise, location-specific investigation steps rather than a complex analytics dashboard.

4. Plant Owner: needs business-level visibility into generation, loss, risk, and potential recovery.

5. Value Proposition

For solar asset operators who need to reduce avoidable generation loss, SolarRecover AI is an operational intelligence platform that connects the PV Digital Twin with AI-based diagnosis and recovery planning, so teams can move from detection to quantified action.

Core value pillars

Pillar	Product value
Understand	Continuously represent the PV asset and its operating context.
Detect	Identify deviations from expected performance.
Explain	Connect anomalies to evidence and probable causes.
Quantify	Estimate energy loss and financial exposure.
Simulate	Compare potential corrective actions and recovery scenarios.
Act	Convert analytics into prioritized maintenance tasks.
Learn	Capture outcomes and improve asset-specific intelligence over time.

6. Product Scope

In scope

- PV asset hierarchy and Digital Twin representation.
- Operational data ingestion and normalization.
- Expected-performance modelling.
- AI/ML anomaly detection.
- Probable root-cause analysis.
- Energy-loss estimation and attribution.
- Financial impact estimation.
- Recovery scenario simulation.
- Maintenance prioritization.
- Context-aware AI Solar Copilot.
- Technician work/feedback capture.

Out of scope for initial MVP

- Replacing full PV design/CAD platforms.
- Replacing SCADA or inverter firmware.
- Autonomous control of live energy infrastructure.
- Manufacturing or installing solar hardware.
- Comprehensive ERP functionality.
- Large-scale autonomous robotics.

7. Core Product Modules

7.1 Universal Solar Data Hub

Accept data from inverter APIs, SCADA exports, smart meters, weather sources, CSV/Excel files, inspection records, and optional imagery. A normalization layer maps heterogeneous inputs into a common asset and telemetry model.

7.2 PV Digital Twin

Represent the physical asset hierarchy: Organization → Site → Plant → Block → Inverter → String → Module → Sensor, while associating weather, production, faults, maintenance, and inspection history.

7.3 Expected Performance Engine

Estimate expected PV output under current conditions using operating history, irradiance, temperature, system configuration, seasonality, and other available variables.

7.4 AI Anomaly Detection

Detect persistent or meaningful deviations using statistical methods, tree-based ML, time-series models, and physics-informed features.

7.5 Root Cause Engine

Generate explainable hypotheses and supporting evidence rather than presenting a black-box fault label.

7.6 Energy Loss Attribution

Estimate the contribution of detected factors such as soiling, shading, inverter issues, degradation, sensor anomalies, and other conditions, while explicitly representing uncertainty.

7.7 Financial Impact Engine

Translate estimated lost generation into configurable financial impact using user-provided tariff, PPA, or market-value assumptions.

7.8 Recovery Simulator

Allow users to test scenarios such as cleaning, inspection, component replacement, or other corrective actions and estimate potential recovered generation, cost, and financial impact.

7.9 Maintenance Priority Engine

Prioritize unresolved issues using explicit factors such as estimated energy loss, risk, downtime, repair cost, warranty state, and potential recovery.

7.10 AI Solar Copilot

Provide natural-language investigation and reporting grounded in the Digital Twin, analytical outputs, and evidence. The LLM is an interface/explanation layer, not the source of numerical truth.

7.11 Technician Mode

Deliver concise issue summaries, inspection checklists, evidence, and post-work recording for field teams.

7.12 Asset Memory & Feedback

Maintain historical faults, interventions, outcomes, and performance changes to support asset-specific intelligence and future model adaptation.

8. Differentiation Strategy

SolarRecover AI should not claim that Digital Twins, AI fault detection, monitoring, inspection analytics, or predictive maintenance are individually new. Research already covers these areas, including PV forecasting, intelligent fault diagnosis, drone imagery, data-driven health-state architectures, and PV O&M; Digital Twin systems. filecite turn5file6 L403-L433

Differentiated product thesis

The differentiation is the closed-loop operational workflow: Digital Twin → diagnosis → energy-loss attribution → financial impact → recovery simulation → maintenance action → repair outcome → model/twin update.

Capability area	Existing landscape	SolarRecover position
PV design/simulation	Mature tools exist	Integrate/consume outputs where useful; not the core battle
OEM monitoring	Strong ecosystem platforms exist	Vendor-neutral intelligence layer
Digital Twin	Research and commercial implementations exist	Operational asset model is the product foundation
Fault detection	Active research and products	Combine detection with explanation and business impact
Inspection analytics	Established in parts of the market	Use as an optional evidence source
Energy-loss attribution	Partial/fragmented	Make it a central product workflow
Recovery simulation	Less commonly integrated end-to-end	Make it a core decision-support capability
Maintenance feedback	Often separated from analytics	Close the loop between action and model history

Positioning statement

SolarRecover AI is positioned as a Solar Energy Recovery Intelligence Layer that can sit above heterogeneous PV data sources and complement existing monitoring, simulation, and inspection systems.

9. Product Experience

Primary navigation

Overview • Assets • Digital Twin • Anomalies • Losses • Recovery • Maintenance • Copilot • Reports • Settings

Executive dashboard

Metric	Example
Plant health	82%
Actual generation	4.82 MW
Expected generation	5.14 MW
Performance gap	-6.2%

Metric	Example
Estimated energy loss	1,248 kWh
Estimated financial impact	₹8,985
Active anomalies	7
High-priority assets	2

Asset investigation flow

Asset → Performance → Expected vs Actual → Anomaly Timeline → Probable Causes → Evidence → Energy Loss → Financial Impact → Recovery Scenarios → Recommended Investigation → Technician Action.

AI Copilot principle

Every consequential answer should expose the analytical evidence behind it. The interface should clearly separate measured values, model estimates, hypotheses, and uncertainty.

10. Technical Architecture

Data sources → Universal Data Layer → PV Digital Twin / Asset Graph → Physics + ML Engine → Root Cause Engine → Loss & Financial Engine → Recovery Simulator → Decision Layer → Users → Feedback

Recommended technology stack

Layer	Recommended technology
Frontend	React + Vite + TypeScript + Tailwind CSS + Recharts/ECharts
Backend	Python + FastAPI + Pydantic + SQLAlchemy
Relational data	PostgreSQL
Time-series	TimescaleDB or PostgreSQL time-series design
Caching / jobs	Redis + background workers
ML	Pandas + NumPy + scikit-learn + XGBoost/LightGBM + PyTorch
AI	LLM API + RAG + structured outputs + tool/function calling
Storage	Object storage for reports and inspection media
Deployment	Containerized cloud deployment with secure API endpoints

AI architecture principle

Numerical prediction, anomaly detection, and loss estimation should come from deterministic/statistical/ML/physics-based services. The LLM should interpret those results, retrieve context, answer questions, and generate reports. This separation reduces the risk of hallucinated numerical claims.

11. Core Data Model

users → organizations → sites → plants → blocks → inverters → strings → modules
Supporting entities: sensors, meters, weather sources, telemetry, production records, anomalies, fault events, diagnoses, loss events, maintenance events, warranty records, simulation scenarios, model

predictions, and feedback.

Digital Twin state model

State	Examples
Physical/current state	Current output, temperature, irradiance, faults, sensor state
Digital/model state	Expected output, health state, risk, degradation estimate
Historical state	Past faults, maintenance, production, recovery outcomes
Relationship state	Plant/block/inverter/string/module dependencies

12. MVP Definition

The MVP should prove one complete operational workflow rather than attempt to reproduce every capability in the long-term vision.

MVP workflow

Upload solar dataset → Configure plant → Analyze → Expected performance → Actual vs expected → AI anomaly detection → Root-cause hypotheses → Energy-loss estimate → Financial impact → Maintenance recommendation → Copilot.

MVP data requirements

- Timestamp
- Power / energy
- Irradiance
- Temperature
- Inverter or asset identifier
- Voltage/current where available
- Plant capacity
- Optional weather and maintenance data

MVP success demonstration

Given historical PV telemetry, the platform should identify a meaningful performance deviation, show the expected-vs-actual trajectory, provide an evidence-backed probable cause, estimate the associated generation loss, convert that estimate into configurable financial impact, and produce an actionable maintenance recommendation.

13. Product Roadmap

Phase	Capabilities
Phase 1 — MVP	CSV ingestion, plant model, expected performance, anomaly detection, loss estimate, financial impact, recommendations, Copilot
Phase 2 — Connected	Real-time APIs, weather integration, temporal models, degradation analytics, maintenance feedback, technician mode
Phase 3 — Intelligent	Self-adaptive modelling, multimodal inspection evidence, warranty intelligence, portfolio analytics

Phase	Capabilities
Phase 4 — Advanced DT	Bidirectional integration where safe, edge intelligence, federated learning, lifecycle intelligence, advanced sustainability

The supplied PVDT review specifically points toward future work on architectures, communication protocols, data management, scalability, transferability, synchronization/latency, cybersecurity, multi-domain modelling, and multi-application implementations. These should guide the research roadmap rather than being forced into the first release. □filecite□turn5file3□L269-L284□

14. Success Metrics

ML / analytical metrics

Capability	Metrics
Expected power	MAE, RMSE, MAPE, R^2
Anomaly detection	Precision, Recall, F1, false-alarm rate
Diagnosis	Precision, Recall, F1, calibrated confidence where appropriate
Forecasting	MAE, RMSE, MAPE, R^2

Operational/product metrics

- Correctly identified loss events.
- Energy loss estimation error.
- Recovery estimate error after confirmed maintenance.
- Mean time to diagnose.
- Mean time to repair.
- False-alarm reduction.
- Percentage of recommendations with traceable evidence.

North-star outcome

Recovered Energy / Avoided Loss — the product should ultimately connect analytics to measurable operational value.

15. Security, Safety & Trust

- Tenant isolation and role-based access control.
- Encrypted data in transit and at rest.
- Authenticated integrations and API keys.
- Audit logging for consequential actions.
- Data validation and anomaly checks at ingestion.
- Model/version tracking and monitoring.
- Backups and recovery procedures.
- Read-only integrations for the early MVP.
- No autonomous control of live PV equipment in the initial release.

Because a production Digital Twin may eventually interact with physical energy assets, safety boundaries should be designed into the architecture from the beginning.

16. Key Risks & Mitigations

Risk	Mitigation
Poor-quality telemetry	Data-quality checks, missingness handling, source confidence, explicit uncertainty
Insufficient fault labels	Semi-supervised/anomaly methods, synthetic scenarios, field validation
False diagnosis	Evidence-based hypotheses, confidence calibration, human confirmation
Model drift	Monitoring, retraining policy, asset-specific baselines
Vendor lock-in	Normalized internal schema and connector architecture
Overclaiming DT capability	Clearly define synchronization and integration maturity
LLM hallucination	Tool-grounded responses, structured outputs, analytical-source citations
Cybersecurity	Least privilege, authentication, encryption, audit trails

17. Competitive Positioning

SolarRecover AI should complement rather than directly replace established categories. PV design/simulation products solve project planning and performance modelling; OEM monitoring products provide equipment-centric operational visibility; inspection/asset-management platforms provide fleet, imagery, and maintenance workflows. SolarRecover's intended position is the decision layer that connects heterogeneous evidence to quantified energy-recovery decisions.

Competitive message

From monitoring solar assets to understanding and recovering lost generation.

18. Product Principles

1. Evidence before explanation. AI recommendations must be traceable to data and model outputs.

2. Digital Twin first. The twin is the operational system model; AI services enrich it.

3. Human-in-the-loop. Recommendations support engineers and technicians rather than silently controlling infrastructure.

4. Vendor-neutral by design. Normalize heterogeneous data instead of designing around one OEM.

5. Uncertainty is a feature. Estimated values must be clearly distinguished from measured values.

6. Business impact matters. Technical anomalies should connect to energy, cost, downtime, and recovery.

7. Close the loop. Post-maintenance outcomes should improve asset history and future intelligence.

8. Build the smallest complete workflow first. MVP scope should prove value before expanding breadth.

19. Final Product Definition

SolarRecover AI is a vendor-neutral Solar Asset Intelligence & Digital Twin platform focused on the operational recovery of lost PV generation. It combines a structured Digital Twin, expected-performance modelling, AI anomaly detection, root-cause reasoning, energy-loss attribution, financial impact

estimation, recovery simulation, maintenance prioritization, technician workflows, and a context-aware AI Copilot.

The core product loop is:

Observe → Model → Detect → Explain → Quantify → Simulate → Act → Measure → Learn

The product's differentiation should be presented as a specific integrated product thesis, not an unsupported claim of absolute global uniqueness. Existing research already demonstrates individual PVDT, forecasting, diagnostic, and maintenance approaches; the opportunity is to build a practical, measurable, vendor-neutral system around the complete recovery decision loop.

Reference basis

Primary research basis: Elnosh, Calais & Parlevliet, A systematic literature review of digital twin research for photovoltaic systems: Trends, challenges, and opportunities, Renewable and Sustainable Energy Reviews (2026), supplied by the user as PVDT.pdf. The review covers 61 peer-reviewed PVDT studies and proposes PVDT eligibility criteria and future research directions. Relevant source excerpts are cited throughout this brief. □filecite□turn5file2□L161-L188□

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